

Phase A endpoint, abstract proof:

$R'_n V = E_{n-1}^+$ for every $H_q(S_n)$ -module V

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19 May 2026

Abstract

We prove the *Phase A endpoint* identity

$$R'_n V = E_{n-1}^+(V)$$

for *every* (finitely generated) module V of the Iwahori–Hecke algebra $H_q(S_n)$ over $\mathbb{Q}(q)$, where $R'_n = (T_{n-1} + 1)(T_{n-2} + 1) \cdots (T_1 + 1)$ and E_{n-1}^+ is the $+q$ -eigenspace of T_{n-1} . More generally, for $W_j := (T_j + 1)(T_{j-1} + 1) \cdots (T_1 + 1)V$ we have $W_j = E_j^+(V)$ for every $1 \leq j \leq n-1$.

The proof is short and does not use the seminormal basis or any shape-specific combinatorics. It rests on two facts: **(i)** T_j and $T_{j'}$ are conjugate in $H_q(S_n)$, so $\dim E_j^+$ is independent of j on any module; **(ii)** the *adjacent disjointness lemma* $E_j^+(V) \cap E_{j+1}^-(V) = 0$, which follows by restriction to the subalgebra $\langle T_j, T_{j+1} \rangle \cong H_q(S_3)$ and a one-line check on each of its three irreducibles.

Consequence. The Phase A endpoint at $\lambda = (2^a, 1^{n-2a})$ for general $a \geq 2$ — the last piece flagged as “mechanical but unwritten” in `2026-05-15-general-a-inductive.tex` (Theorem 2.1) and `2026-05-18-final-pair-identification.tex` (Section 6 “What this paper does NOT close”) — is now proved unconditionally. Combined with the May-15 inductive reduction and the May-18 final pair identification, this closes the rank-zero theorem $\Pi^{S_n}|_{V_{(2^a, 1^{n-2a})}} = 0$ for every $a \geq 1$, $n \geq 3a$ **unconditionally**.

1 Setup and notation

Let $H_q(S_n)$ denote the Iwahori–Hecke algebra over the field $\mathbb{Q}(q)$, with generators $T_1, \dots, T_{n-1}$ subject to the quadratic relations $T_i^2 = (q-1)T_i + q$ and the braid relations

$$T_i T_{i+1} T_i = T_{i+1} T_i T_{i+1} \quad (1 \leq i \leq n-2), \quad T_i T_j = T_j T_i \quad (|i-j| \geq 2).$$

Over $\mathbb{Q}(q)$ (with q transcendental), $H_q(S_n)$ is semisimple, with irreducible representations V_λ indexed by partitions $\lambda \vdash n$.

For an $H_q(S_n)$ -module V and an index $1 \leq j \leq n-1$ define the T_j -eigenspaces

$$E_j^+(V) := \ker(T_j - q \cdot \text{id})|_V, \quad E_j^-(V) := \ker(T_j + \text{id})|_V.$$

Since $T_j^2 = (q-1)T_j + q$, the minimal polynomial of T_j divides $(X-q)(X+1)$, hence T_j acts diagonalizably on V with $V = E_j^+(V) \oplus E_j^-(V)$. We write $\pi_j^+ : V \rightarrow V$ for the projection onto E_j^+ along E_j^- , so $\pi_j^+ + \pi_j^- = \text{id}$ and the operator identity

$$T_j + 1 = (1+q) \pi_j^+ \tag{1}$$

holds: indeed, $T_j + 1$ vanishes on E_j^- and acts as $(1 + q)$ on E_j^+ .

Write the staircase factor

$$R'_k := (T_{k-1} + 1)(T_{k-2} + 1) \cdots (T_1 + 1)$$

and the partial product

$$W_j(V) := (T_j + 1)(T_{j-1} + 1) \cdots (T_1 + 1)V = (T_j + 1)W_{j-1}(V),$$

with $W_0(V) := V$. Note $W_{n-1}(V) = R'_n V$.

2 The adjacent disjointness lemma

The single substantive ingredient is:

Lemma 1 (Adjacent disjointness). *Let V be any $H_q(S_n)$ -module over $\mathbb{Q}(q)$. For every $1 \leq j \leq n-2$,*

$$E_j^+(V) \cap E_{j+1}^-(V) = 0.$$

Symmetrically, $E_j^-(V) \cap E_{j+1}^+(V) = 0$ for every $1 \leq j \leq n-2$.

Proof. The two generators T_j, T_{j+1} satisfy the braid relation $T_j T_{j+1} T_j = T_{j+1} T_j T_{j+1}$ and the quadratic relations $T_j^2 = (q-1)T_j + q$, $T_{j+1}^2 = (q-1)T_{j+1} + q$. Hence the unital subalgebra $A_j := \langle T_j, T_{j+1} \rangle \subseteq H_q(S_n)$ is a quotient of $H_q(S_3)$ (with $T_j \mapsto T_1$, $T_{j+1} \mapsto T_2$ on the generators of $H_q(S_3)$). Since $H_q(S_3)$ is semisimple of dimension 6, and A_j contains two non-commuting elements of the same algebra, A_j is in fact isomorphic to $H_q(S_3)$, but we do not need this — we only need that A_j is a quotient of $H_q(S_3)$, hence every A_j -module restricts from $H_q(S_3)$ along the quotient.

Restrict V to A_j . Since $H_q(S_3)$ is semisimple over $\mathbb{Q}(q)$, so is its quotient A_j , so $V|_{A_j}$ decomposes as a direct sum of irreducible A_j -modules, each of which is an irreducible $H_q(S_3)$ -module pulled back along A_j .

The irreducibles of $H_q(S_3)$ over $\mathbb{Q}(q)$ are:

- **Trivial** $V_{(3)}$: $\dim = 1$, $T_1 = T_2 = q$.
- **Sign** $V_{(1^3)}$: $\dim = 1$, $T_1 = T_2 = -1$.
- **Standard** $V_{(2,1)}$: $\dim = 2$.

For the standard rep, we use the seminormal basis indexed by the two SYTs of shape $(2, 1)$: T_A (rows $\{1, 2\}, \{3\}$) and T_B (rows $\{1, 3\}, \{2\}$). In this basis $T_1 = \text{diag}(q, -1)$ (since $1, 2$ are same-row in T_A and same-column in T_B). The generator T_2 acts as a 2×2 non-diagonal Hecke block: T_2 couples v_{T_A} and v_{T_B} (axial distance ± 2) because $2, 3$ are in different rows and columns of both SYTs. Concretely, $T_2 v_{T_A} = \alpha v_{T_A} + \beta v_{T_B}$ with $\beta \neq 0$.

We now verify the lemma on each irreducible:

Trivial rep. $E_{T_2}^- = \ker(T_2 + 1) = \ker((q+1) \cdot \text{id}) = 0$ since $q+1 \neq 0$ over $\mathbb{Q}(q)$. Hence $E_{T_1}^+ \cap E_{T_2}^- = 0$. Similarly $E_{T_1}^- = 0$, so $E_{T_1}^- \cap E_{T_2}^+ = 0$.

Sign rep. $E_{T_1}^+ = \ker((-1) - q \cdot \text{id}) = \ker(-(q+1) \cdot \text{id}) = 0$. Hence $E_{T_1}^+ \cap E_{T_2}^- = 0$. Symmetrically, $E_{T_2}^+ = 0$, so $E_{T_1}^- \cap E_{T_2}^+ = 0$.

Standard rep. $E_{T_1}^+ = \text{span}(v_{T_A})$ is one-dimensional, spanned by v_{T_A} . Suppose $v_{T_A} \in E_{T_2}^-$, i.e., $T_2 v_{T_A} = -v_{T_A}$. But $T_2 v_{T_A} = \alpha v_{T_A} + \beta v_{T_B}$ with $\beta \neq 0$, so the v_{T_B} -component on the left is nonzero while on $-v_{T_A}$ it is zero, a contradiction. Therefore $E_{T_1}^+ \cap E_{T_2}^- = 0$.

To see the symmetric statement $E_{T_1}^- \cap E_{T_2}^+ = 0$: $E_{T_1}^- = \text{span}(v_{T_B})$ is one-dimensional. Suppose $v_{T_B} \in E_{T_2}^+$, i.e., $T_2 v_{T_B} = q v_{T_B}$. T_2 is a non-degenerate 2-block, so $T_2 v_{T_B}$ has nonzero v_{T_A} -component (mirror of the previous argument with the roles of T_A, T_B swapped via the axial-distance sign change). Same contradiction. Therefore $E_{T_1}^- \cap E_{T_2}^+ = 0$.

The lemma now follows: $V|_{A_j}$ is a direct sum of irreducible A_j -modules, each of which is one of the three $H_q(S_3)$ -irreps above (via the quotient $H_q(S_3) \twoheadrightarrow A_j$). $E_j^+(V) \cap E_{j+1}^-(V)$ commutes with the decomposition (it is the intersection of two A_j -stable subspaces in $V|_{A_j}$), and each summand contributes 0. $\square$

Remark 2 (Why “standard” suffices for off-diagonal blocks). The argument for the standard rep shows that in any module V , within any non-degenerate 2-block of T_2 (in the $\langle T_1, T_2 \rangle$ -restriction structure), the T_1 -eigenstates are not T_2 -eigenstates. The trivial and sign reps are degenerate as A_j -modules: T_1 and T_2 act as the same scalar. In these degenerate cases the eigenspace intersections vanish trivially for parity reasons. The non-trivial content of the lemma lives entirely in the 2-dimensional standard rep, where the off-diagonal Hoefsmit coupling forces the eigenstate mismatch.

3 Trace conjugacy and the dimension constancy

Lemma 3 (Trace conjugacy). *For any $H_q(S_n)$ -module V and $1 \leq j, j' \leq n-1$, $\dim E_j^+(V) = \dim E_{j'}^+(V)$ and $\dim E_j^-(V) = \dim E_{j'}^-(V)$.*

Proof. Using the braid relation $T_j T_{j+1} T_j = T_{j+1} T_j T_{j+1}$, let $h := T_j T_{j+1}$. Then in $H_q(S_n)$,

$$h \cdot T_j \cdot h^{-1} = (T_j T_{j+1}) T_j (T_j T_{j+1})^{-1} = (T_j T_{j+1} T_j) T_{j+1}^{-1} T_j^{-1} = (T_{j+1} T_j T_{j+1}) T_{j+1}^{-1} T_j^{-1} = T_{j+1}.$$

Hence T_j and T_{j+1} are conjugate elements of $H_q(S_n)$. By induction, T_j and $T_{j'}$ are conjugate for any $1 \leq j, j' \leq n-1$.

Conjugate elements have the same trace on any module V : $\text{tr}(T_j|_V) = \text{tr}(T_{j'}|_V)$. Combined with T_j 's diagonalizability with eigenvalues $\{q, -1\}$:

$$\text{tr}(T_j|_V) = q \dim E_j^+(V) - \dim E_j^-(V), \quad \dim E_j^+(V) + \dim E_j^-(V) = \dim V.$$

Solving for $\dim E_j^+(V)$ in terms of $\dim V$ and $\text{tr}(T_j|_V)$ shows the former depends only on V , not on j . $\square$

4 Phase A endpoint

Theorem 4 (Phase A endpoint, general). *Let V be any $H_q(S_n)$ -module over $\mathbb{Q}(q)$. Then*

$$W_j(V) = E_j^+(V) \quad \text{for every } 1 \leq j \leq n-1. \quad (2)$$

In particular, $R_n^l V = W_{n-1}(V) = E_{n-1}^+(V)$.

Proof. By induction on j .

Base case $j = 1$. $W_1(V) = (T_1 + 1)V = (1 + q)\pi_1^+(V) = (1 + q)E_1^+(V) = E_1^+(V)$ (since $(1 + q) \in \mathbb{Q}(q)^\times$).

Inductive step $j - 1 \rightarrow j$. Assume $W_{j-1}(V) = E_{j-1}^+(V)$. Then

$$W_j(V) = (T_j + 1)W_{j-1}(V) = (T_j + 1)E_{j-1}^+(V) = (1 + q)\pi_j^+(E_{j-1}^+(V)) = \pi_j^+(E_{j-1}^+(V)),$$

where we used (1) and the scalar invertibility of $(1 + q)$.

It remains to identify $\pi_j^+(E_{j-1}^+(V))$ with $E_j^+(V)$. The kernel of $\pi_j^+|_{E_{j-1}^+(V)}$ is

$$E_{j-1}^+(V) \cap \ker(\pi_j^+) = E_{j-1}^+(V) \cap E_j^-(V) = 0$$

by Lemma 1. Hence $\pi_j^+|_{E_{j-1}^+(V)}$ is injective, and

$$\dim \pi_j^+(E_{j-1}^+(V)) = \dim E_{j-1}^+(V) = \dim E_j^+(V)$$

by Lemma 3. Since $\pi_j^+(E_{j-1}^+(V)) \subseteq E_j^+(V)$ and the two have the same finite dimension, they are equal. Hence $W_j(V) = E_j^+(V)$. $\square$

Corollary 5 (Iterated Phase A: image of R'_n). *For every V , $R'_n V = E_{n-1}^+(V)$ as subspaces of V .*

Remark 6. The proof of Theorem 4 makes no reference to the seminormal basis, to specific partitions, or to any combinatorial description of V . In particular:

- It applies uniformly to *every* V_λ .
- It gives a one-line description of the image (an eigenspace), abstracting away the basis-tracking machinery of May-13/14.
- It works for arbitrary V , including direct sums, allowing the result to be used inductively in restriction arguments.

5 Consequences for $\lambda = (2^a, 1^{n-2a})$ and the rank-zero theorem

The Phase A endpoint at general $a \geq 2$, $n \geq 3a$ was the last ingredient flagged as “mechanical but unwritten” in two prior papers:

- 2026-05-15-general-a-inductive.tex Theorem 2.1 (Phase A endpoint at general a) — proved for $a = 2, 3$ via basis $\mathcal{B}_j$; for $a \geq 4$ “mechanical generalization not written out.”
- 2026-05-18-final-pair-identification.tex Section 6, “What this paper does NOT close,” item 1: “Phase A at general a : the orthogonal piece. Was conditional in May-17, remains conditional today. The structural argument at $a = 2, 3$ is concrete; for $a \geq 4$ it’s a routine generalization I haven’t sat down to formalize.”

Theorem 4 applied with $V = V_\lambda$ for $\lambda = (2^a, 1^{n-2a})$ gives this endpoint immediately, for every $a \geq 1$ and every n :

$$R'_n V_{(2^a, 1^{n-2a})} = E_{n-1}^+|_{V_{(2^a, 1^{n-2a})}}.$$

Theorem 7 (Rank-zero theorem for $(2^a, 1^{n-2a})$, unconditional). *For every $a \geq 1$ and every $n \geq 3a$,*

$$\Pi^{S_n}|_{V_{(2^a, 1^{n-2a})}} = 0.$$

Proof. The chain of implications, with Phase A endpoint now unconditional:

1. Phase A endpoint (Theorem 4 with $V = V_{(2^a, 1^{n-2a})}$): $R'_n V_\lambda = E_{n-1}^+|_{V_\lambda}$.
2. May-15 inductive reduction (2026-05-15-general-a-inductive.tex Theorem 4.1): combined with Phase A endpoint at λ ,

$$B_{n-a+1}^{(\lambda)} V_\lambda = (T_{n-a} + 1)(T_{n-a+1} + 1) \cdots (T_{n-2} + 1) \Phi_1(B_{n-a}^{(\mu_1)} V_{\mu_1}),$$

where $\mu_1 = (2^{a-1}, 1^{n-2a})$ and Φ_1 is the branching isomorphism (May-15 Definition 3.1).

3. May-17/18 per-trajectory framework (2026-05-18-final-pair-identification.tex Theorem 1.1): the right-hand side is one-dimensional with support exactly $S_a^{(n)}$ ($|S_a^{(n)}| = 2^a$), all coefficients nonzero in $\mathbb{Q}(q)$. Every SYT in $S_a^{(n)}$ has its column-2 row-1 entry $\geq n - 2a + 1 \geq a + 1 \geq 3$, hence lies in E_1^- (via the column-1 position of 2 in the SYT, since $a_1 = 2$ is not in the support).
4. Therefore $B_{n-a+1}^{(\lambda)} V_\lambda \subseteq E_1^-$, and the factorization $\Pi^{S_n} = R'_2 \cdots R'_{n-a} \cdot B_{n-a+1}^{(\lambda)}$ has R'_{n-a} ending in $(T_1 + 1)$, which kills E_1^- . Hence $\Pi^{S_n} V_\lambda = 0$.

□

Remark 8 (Status table). With Theorem 7, the rank-zero theorem for Π^{S_n} now has unconditional structural proofs for the following infinite families:

Family	Status	Reference
$\lambda = (1^n)$ (sign)	trivial (every $(T_i + 1)$ kills)	—
$\lambda = (2, 1^{n-2})$	May-11, hook $k = 2$	2026-05-11-j-formula.tex
$\lambda = (3, 1^{n-3})$	May-12, hook $k = 3$	2026-05-12-hook-j-formula.tex
$\lambda = (2, 2, 1^{n-4})$	May-13, first non-hook	2026-05-13-non-hook-22-1n.tex
$\lambda = (2^a, 1^{n-2a})$, $a \geq 2$, $n \geq 3a$	May-19 (this paper)	this writeup + May-15, May-18

6 Computational verification

We verified Lemma 1 and Theorem 4 computationally at $q = 7 \in \mathbb{Q}$ (generic for the seminormal basis) on a representative set of partitions.

Verification of Lemma 1: $E_j^+ \cap E_{j+1}^- = 0$ for all $j \in [1, n-2]$.

λ	n	$\dim V_\lambda$	verified
$(2, 1)$	3	2	PASS
$(3, 1), (2, 2), (2, 1, 1)$	4	≤ 3	PASS
$(3, 1, 1), (2, 2, 1), (3, 2), (2, 1^3)$	5	≤ 6	PASS
$(2, 2, 1, 1)$	6	9	PASS
$(2, 2, 1, 1, 1)$	7	14	PASS
$(2^3, 1^3)$	9	48	PASS
$(2^3, 1^4)$	10	90	PASS
$(2^4, 1^4)$	12	297	PASS
$(3, 3), (3, 2, 1), (4, 2, 1)$	≤ 7	≤ 35	PASS

Verification of Theorem 4: $R'_n V_\lambda = E_{n-1}^+$.

λ	$\dim R'_n V_\lambda$	$\dim E_{n-1}^+$
$(2, 2, 1, 1)$ ($a = 2, n = 6$)	3	3
$(2, 2, 1, 1, 1)$ ($a = 2, n = 7$)	4	4
$(2^3, 1^3)$ ($a = 3, n = 9$)	14	14
$(2^3, 1^4)$ ($a = 3, n = 10$)	20	20
$(2^4, 1^4)$ ($a = 4, n = 12$)	75	75
$(3, 2, 1)$ ($n = 6$)	8	8
$(3, 1^4)$ ($n = 7$)	5	5

Script: `~/projects/scratch/2026-05-19-phase-a-general/verify_phase_a.py`.

7 Discussion

7.1 Why this proof works and prior proofs didn't try

The prior Phase A proofs at $a = 2$ (May-13) and $a = 3$ (May-14) used *explicit* basis-tracking through $\mathcal{B}_j$, a set of $T_j + q$ -eigenvectors built from the seminormal basis. This approach has two virtues:

- It directly produces an explicit basis of $W_j = E_j^+$ at every step, useful for the subsequent Phase B analysis.
- It is shape-aware: the basis encodes the column-1 corners (rows $\geq a + 1$) and column-2 corner (row a) of λ explicitly.

But it has a cost: the case analysis at each $(T_j + 1)$ step involves $O(a)$ sub-cases per row, and the combinatorial bookkeeping does not naturally telescope. Generalizing to $a \geq 4$ required re-doing the case analysis with an extra column-2 row, hence “mechanical but unwritten.”

The abstract proof here drops the explicit basis and uses only: (i) $W_j = \pi_j^+(W_{j-1})$ (an operator identity); (ii) trace conjugacy (a Hecke-algebra identity); (iii) adjacent disjointness $E_j^+ \cap E_{j+1}^- = 0$ (a Hecke identity for the subalgebra generated by two adjacent generators).

None of these depend on λ or on the specific shape's combinatorics. The shape-information enters only through the *starting point* V , and exits only through the *endpoint* $E_{n-1}^+(V)$.

7.2 The lemma in the larger picture

The adjacent disjointness lemma $E_j^+ \cap E_{j+1}^- = 0$ is a crisp algebraic statement about the relative geometry of T_j -eigenspaces in any $H_q(S_n)$ -module. It says: the $+q$ -eigenspace of T_j and the -1 -eigenspace of T_{j+1} are in *general position* in the strongest sense (they meet only at the origin).

Equivalent statements:

- The natural map $E_j^+ \rightarrow V/E_{j+1}^- \cong E_{j+1}^+$ (projection along E_{j+1}^-) is injective.
- The eigenstates of T_j “rotate fully” under restriction to $\langle T_j, T_{j+1} \rangle$: no $T_j + q$ -eigenstate is simultaneously a $T_{j+1} - 1$ -eigenstate.
- Equivalently (by dimension count): $E_j^+ + E_{j+1}^- = V$ (the two adjacent generic-position subspaces span V).

For irreducibles, this rotation is essentially the content of the non-degeneracy of the off-diagonal block in the 2×2 Hoefsmit form for the $H_q(S_3)$ -restriction.

7.3 What this does not give us

Theorem 4 gives Phase A as a set-theoretic identity (“ $W_{n-1} = E_{n-1}^+$ ”), but does not produce an explicit basis of W_{n-1} . For applications that require an explicit identification of the image (such as the inductive reduction formula in `2026-05-15-general-a-inductive.tex`), one combines this theorem with the seminormal-basis identification of E_{n-1}^+ on $V_{(2^a, 1^{n-2a})}$ with a copy of V_{μ_1} on $n-2$ letters via the branching iso Φ_1 .

Phase A endpoint here serves as the foundation; the subsequent phases (Phase A₂, *A₃ in the May-14/May-15 framework*) need shape – specific input to track which 2^a SYTs end up in the support, and those steps remain the proper subject of the May-17 / May-18 trajectory analysis.

7.4 Generality of the proof

We emphasize: Theorem 4 holds for *any* module V , with no restriction on λ , dimension, or generic position. In particular it also gives Phase A endpoint for the hook families $\lambda = (k, 1^{n-k})$, the symmetric partitions $(n-1, 1)$, etc. In every case $R'_n V_\lambda = E_{n-1}^+|_{V_\lambda}$ holds. Many of the intermediate inductive lemmas in our prior writeups can now be stated cleanly using this fact instead of re-deriving Phase A from basis tracking.

8 Honest assessment

What this paper achieves.

1. Theorem 4: Phase A endpoint $W_j(V) = E_j^+(V)$ for every $H_q(S_n)$ -module V and every $1 \leq j \leq n-1$, proved unconditionally.
2. Lemma 1 (adjacent disjointness): the underlying Hecke identity, proved via $H_q(S_3)$ -restriction and a 3-case check.
3. Theorem 7: as a consequence, the rank-zero theorem for the family $\lambda = (2^a, 1^{n-2a})$ is now *unconditional* for every $a \geq 1$, $n \geq 3a$, eliminating the last conditional ingredient flagged in May-18.

What this paper does not address.

1. The optimal-threshold question $n = 3a$. Theorem 4 holds without any threshold; the threshold $n \geq 3a$ enters in the support analysis (May-18), not in Phase A. Tightness of $n = 3a$ remains open.
2. Other rank-zero families. The next natural family $(3, 2, 1^{n-5})$ (where the j -formula at $j = \ell+1$ does *not* immediately collapse to dim 1, per PROVE.md backup target) is unaffected.
3. An explicit description of the image basis. As discussed, the theorem gives a set-theoretic identity but not a constructive basis. For most applications the trivial basis from the seminormal identification suffices.

Why this proof was not found earlier. The proofs at $a = 2$ (May-13) and $a = 3$ (May-14) used basis tracking and worked. The natural inclination was to generalize the same proof to $a \geq 4$ — a mechanical task but tedious enough to be deferred across multiple sessions. The abstract version comes from stepping back and asking “what does the basis tracking actually prove?” The answer is “ $\pi_j^+(W_{j-1}) = E_j^+$,” which is shape-independent. Once written this way, the proof reduces to a one-line dimension count plus a one-page check at $H_q(S_3)$.

This is a recurring pattern in our work on the $\lambda = (2^a, 1^{n-2a})$ family: the May-13 proof was explicit and shape-specific; May-15 abstracted it via the branching iso; May-17 abstracted further via the per-trajectory invariant; May-18 identified the final pair via closed-form S_a . Each round of abstraction shed shape-specificity. The present paper is the final layer of abstraction *for Phase A*, reducing it to a generic Hecke statement that does not see the partition at all.